

Lower-Division Requirements

I&C SCI 31: Introduction to Programming

- **Direct Prerequisites Required:** None (or AP Computer Science A score of 3+ for credit)
- **Minimum Grade:** C or better to advance to ICS 32
- **Is Prerequisite For:** ICS 32

I&C SCI 32: Programming with Software Libraries

- **Direct Prerequisites Required:**
 - I&C SCI 31 (Minimum grade of C)
- **Is Prerequisite For:** ICS 33

I&C SCI 33: Intermediate Programming

- **Direct Prerequisites Required:**
 - I&C SCI 32 (Minimum grade of C)
- **Is Prerequisite For:** ICS 45C, ICS 46, SWE 43

I&C SCI 45C: Programming in C/C++ as a Second Language

- **Direct Prerequisites Required:**
 - I&C SCI 33 (Minimum grade of C)
- **Is Prerequisite For:** ICS 46, ICS 51, CS 122A

I&C SCI 46: Data Structure Implementation and Analysis

- **Direct Prerequisites Required:**
 - I&C SCI 33 (Minimum grade of C)
 - I&C SCI 45C (Minimum grade of C)
 - I&C SCI 6B (Boolean Logic and Discrete Structures)
- **Is Prerequisite For:** CS 161, CS 122A, CS 143A, CS 171, SWE 117

I&C SCI 51: Introductory Computer Organization

- **Direct Prerequisites Required:**
 - I&C SCI 33
 - I&C SCI 45C
 - I&C SCI 6B
- **Is Prerequisite For:** ICS 53, CS 152

I&C SCI 53: Principles in System Design

- **Direct Prerequisites Required:**
 - I&C SCI 51
 - I&C SCI 46
- **Is Prerequisite For:** CS 143A, CS 132

I&C SCI 6B: Boolean Logic and Discrete Structures

- **Direct Prerequisites Required:** None
- **Is Prerequisite For:** ICS 6D, ICS 46, ICS 51

I&C SCI 6D: Discrete Mathematics for Computer Science

- **Direct Prerequisites Required:**
 - I&C SCI 6B
- **Is Prerequisite For:** CS 161, CS 162

MATH 2A: Single-Variable Calculus I

- **Direct Prerequisites Required:** Satisfactory placement score or MATH 1B
- **Is Prerequisite For:** MATH 2B

MATH 2B: Single-Variable Calculus II

- **Direct Prerequisites Required:**
 - MATH 2A
- **Is Prerequisite For:** ICS 6N / MATH 3A, STATS 67

I&C SCI 6N / MATH 3A: Computational Linear Algebra

- **Direct Prerequisites Required:**
 - MATH 2B
- **Is Prerequisite For:** CS 178, CS 112, CS 116

STATS 67: Introduction to Probability and Statistics for CS

- **Direct Prerequisites Required:**
 - MATH 2B
- **Is Prerequisite For:** CS 161, CS 178, CS 171

Upper-Division CS Core & Electives

COMPSCI 161: Design and Analysis of Algorithms

- **Direct Prerequisites Required:**
 - I&C SCI 46
 - I&C SCI 6D
 - MATH 2B
 - STATS 67 (or MATH 130A)
- **Is Prerequisite For:** CS 162, CS 163, CS 164, CS 177

COMPSCI 122A: Introduction to Data Management

- **Direct Prerequisites Required:**
 - I&C SCI 45C or I&C SCI 46
- **Is Prerequisite For:** CS 122B (Project in Data Management)

COMPSCI 143A: Principles of Operating Systems

- **Direct Prerequisites Required:**
 - I&C SCI 53
 - I&C SCI 46
- **Is Prerequisite For:** CS 143B (Project in Operating Systems)

COMPSCI 171: Introduction to Artificial Intelligence

- **Direct Prerequisites Required:**
 - I&C SCI 46
 - STATS 67 (or MATH 130A)
- **Is Prerequisite For:** CS 175, CS 178

COMPSCI 178: Machine Learning and Data Mining

- **Direct Prerequisites Required:**
 - I&C SCI 6N (or MATH 3A)
 - STATS 67
 - I&C SCI 46
- **Is Prerequisite For:** Advanced ML electives